

A modular, multi-GPU training pipeline for low-resource cross-lingual speaker verification: design, ablation, and reference deployment on Sinhala–Tamil bilingual audio

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Abstract

We describe the design, implementation, and empirical validation of an open-source training pipeline for cross-lingual speaker verification under low-resource constraints. The pipeline extends the upstream `voxceleb_trainer` codebase with eleven modular features — each selectable via a single configuration key — spanning multilingual self-supervised front-ends, cross-lingual fine-tuning with layer-wise learning-rate decay, auxiliary language-identification heads, score normalisation, and PLDA scoring. We deploy the pipeline across a heterogeneous cluster of NVIDIA T4, A10, and A40 GPUs, with distributed data-parallel training on the T4 pair and single-GPU runs for ablation passes. As a reference deployment, we evaluate on a two-tier Sinhala–Tamil benchmark: v0, 527 speakers of read speech from OpenSLR SLR52 and SLR65 (478 Sinhala, 49 Tamil; 12,000 trials per language), and v1, built on SLCeleb, our 280-speaker in-the-wild corpus (pending data placement), with single-feature ablations isolating contributions. The toolkit, master experimental protocol, corpus-preparation script, and aggregation tools are released to enable reproducible low-resource SV research.

Keywords: speaker verification; cross-lingual transfer; distributed training; multi-GPU; open-source toolkit; low-resource languages; ablation study.

1 Introduction

Modern speaker verification (SV) systems are increasingly defined as much by the engineering of the training pipeline as by the choice of architecture or loss function. A typical research setup combines several independently-developed components: a front-end (mel / SincConv / self-supervised), a backbone (TDNN / ResNet / transformer), a margin-based loss (AM- / AAM-Softmax), an evaluation backend (cosine / PLDA), score normalisation, and one or more auxiliary training objectives. Yet most published systems combine these components in a single end-to-end commitment, without ablating contributions, and without releasing the configuration scaffolding that would let a downstream practitioner enable or disable each lever.

This paper presents the inverse: a training pipeline that decomposes the SV system into eleven *independently toggable* features (FEATURE-001 through FEATURE-011 in the codebase), each guarded by a single YAML or CLI key, each accompanied by a design document, and each contributing to a unified single-feature ablation. We describe the architectural decisions that make this modularity work in practice (Section 3), the heterogeneous multi-GPU deployment we use to run the protocol (Section 4), and the empirical validation on a low-resource Sinhala–Tamil corpus (Section 5). Contributions:

1. A modular SV training pipeline with eleven independently toggable features, each composable with all others and each defaulting to off so that pre-existing experiment scripts remain behaviour-identical.

2. A heterogeneous multi-GPU deployment strategy that maps each protocol stage to the appropriate device class (DDP on consumer GPUs, single-GPU for ablations, large-VRAM for self-supervised backbones).
3. A reproducible two-tier Sinhala–Tamil benchmark (v0: 527 speakers from OpenSLR SLR52/SLR65 read speech; v1: SLCeleb, our 280-speaker in-the-wild corpus, pending) as the pipeline’s reference deployment, with corpus preparation and results aggregation fully automated.

2 Related work

2.1 Speaker verification architectures

ECAPA-TDNN [4] is the de-facto reference architecture for non-self-supervised SV; multilingual self-supervised encoders (WavLM [2], wav2vec 2 [1]) provide strong out-of-the-box features for low-resource languages. Margin-based losses (AAM-Softmax [3]) are standard for embedding training.

2.2 Cross-lingual transfer and SV-specific eval backends

Cross-lingual fine-tuning from English-pretrained checkpoints is the established remedy for low-resource SV. Eval-time backends such as PLDA [7] and AS-Norm [6] add 5–20% relative MinDCF reduction at zero training cost. Adversarial language/channel invariance [5] is infrastructure-ready in our pipeline but evaluated only as future work in this paper.

2.3 Modular training pipelines in the broader AI ecosystem

Modular training frameworks — HuggingFace `transformers`, SpeechBrain, NeMo, fairseq — have shaped the way modern ML research is conducted. Our pipeline shares their feature-toggle philosophy but specialises it for SV: each feature is paired with a design document articulating the hypothesis, expected effect size, and risk; each is ablated explicitly in a single command sweep.

3 System design

3.1 Feature toggle as the primary abstraction

Every shipped feature obeys a strict invariant: *the default-off configuration is byte-identical to the pre-feature pipeline*. This constraint — enforced by design review for every merge — means a practitioner can adopt the pipeline incrementally, enabling one feature at a time, with confidence that earlier experiments remain valid.

Concretely:

- Every feature is gated by a single boolean (e.g. `plda: true`); the YAML and CLI representations are isomorphic.
- Default-off paths execute the original code path with no additional branching cost.
- Each feature ships with a smoke test verifying behaviour parity at the default setting.

3.2 Composition rules

The eleven features compose into the following layered architecture (Figure ??, [INSERT_FIGURE]):

Table 1: Heterogeneous-cluster GPU allocation for the protocol.

Machine	VRAM	Workload
A40 (single node)	48 GB	Self-supervised front-end configs (FEATURE-001 WavLM / XLS-R / mHuBERT)
2× T4 (DDP)	16 GB × 2	ECAPA-TDNN baseline, primary protocol; distributed data-parallel
A10 (single node)	24 GB	Single-feature ablations, multi-seed reproducibility runs

Front-end

Mel-filterbank, SincConv (raw-waveform), or self-supervised transformer (WavLM / XLS-R / mHuBERT-147).

Backbone

ECAPA-TDNN, ResNet-SE34V2, RawNet3, MLPMixerspeaker (raw or mel).

Pooling

Channel-attentive statistical pooling (ECAPA), self-attentive pooling (others).

Training-time auxiliary heads

Multi-task language-ID, DANN-style adversarial language/channel.

Loss

AAM-Softmax, AM-Softmax, ProtoNet, GE2E, Triplet.

Optimisation

Selective freezing, layer-wise LR decay, distributed data-parallel.

Eval backends

Cosine, PLDA (two-covariance), AS-Norm (composable with both).

Reporting

Per-language EER / MinDCF, pooled metrics, multi-seed aggregation.

3.3 Configuration management and reproducibility

A single YAML config encodes every choice in the pipeline. Paths expand environment variables (`{SL_SPV_DATA_ROOT}`) for machine portability. The deterministic-mode switch (`deterministic: true`) sets cuDNN deterministic flag, disables TF32, and enables `torch.use_deterministic_algorithms` for paper-grade reproducibility.

4 Multi-GPU deployment

4.1 Heterogeneous cluster allocation

The reference deployment uses three machine types in a small heterogeneous cluster:

Distributed data-parallel on the T4 pair uses NCCL backend; the trainer reuses the upstream `voxceleb_trainer` DDP path (verified after BUGFIX-001 corrected the original `kwarg-passing` bug in `mp.spawn`).

Table 2: Preliminary EER (%) on benchmark v0 (527 read-speech speakers; 12,000 trials per language). Fine-tuned rows are mean $\pm$ std over three seeds. Preliminary, benchmark v0 (read speech, closed-set for the fine-tuned rows); distillation and on-device results are pending.

System	Params	Si	Ta
Zero-shot SpeechBrain ECAPA-TDNN	$\sim 20\text{M}$	4.78	3.21
Zero-shot ReDimNet-B1	2.2M	3.57	1.67
Zero-shot ReDimNet-B6	15M	2.71	1.48
ReDimNet-B6 + AS-Norm (training-free)	15M	2.07	0.90
WavLM-base+ LoRA fine-tune (3 seeds)	95M (1.1% trained)	3.73 ± 0.49	3.11 ± 0.61
WavLM-base+ full fine-tune (3 seeds)	95M	1.69 ± 0.06	1.41 ± 0.10

4.2 Wall-clock budget

The full protocol (three primary configurations $\times$ three seeds + five single-feature ablations $\times$ one seed) dispatches to $\sim 3\text{--}5$ days of wall-clock time across the three machines.

5 Experiments

5.1 Reference deployment: two-tier Sinhala–Tamil benchmark

The reference deployment is a two-tier benchmark. Benchmark v0 comprises 527 speakers of read speech: 478 Sinhala from OpenSLR SLR52 and 49 Tamil from OpenSLR SLR65 (speakers with at least 10 utterances). Benchmark v1 is built on SLCeleb, our 280-speaker in-the-wild Sinhala–Tamil corpus (multi-genre, multi-session), and is pending data placement. Train/test split is 80/20 within each speaker; trial pairs (v0: 3,000 target + 9,000 impostor pairs per language) are generated deterministically from a single random seed (`tools/sl_dataprep.py`). Per-language, cross-lingual, and pooled trial lists are evaluated independently; the cross-lingual list is empty at v0 (no bilingual speakers in OpenSLR) and is populated at v1 by SLCeleb.

5.2 Configurations

- P0: ECAPA-TDNN from scratch.
- P1: P0 + cross-lingual fine-tune from VoxCeleb1 ECAPA checkpoint, front-end frozen, layer-wise LR decay $\gamma = 0.9$.
- P2: P1 + multi-task language-ID head ($\lambda = 0.3$) + AS-Norm (top- $K = 300$, 500-speaker cohort) + PLDA ($d = 200$) + per-language evaluation.

5.3 Preliminary results (benchmark v0)

Table 2 gives preliminary benchmark-v0 context for the edge study: the 2.2M-parameter ReDimNet-B1 already outperforms the $\sim 20\text{M}$ ECAPA-TDNN zero-shot, AS-Norm is a training-free lever worth carrying on-device (Si $2.71 \rightarrow 2.07$), and the fine-tuned WavLM-base+ (full fine-tune) is the current distillation-teacher candidate. Per-language score calibration proved mandatory (cross-language calibrator swap inflates Cllr up to $5.7\times$) — an on-device deployment must ship per-language calibrators. The distillation, quantisation, and runtime results that are this paper’s subject remain pending (P6).

Table 3: Equal-error rate (%) on the SL-SPV evaluation set, mean $\pm$ std over three random seeds.

Config	Pooled	Si	Ta	Cs
P0 from scratch	[X.X $\pm$ Y.Y]	[X.X $\pm$ Y.Y]	[X.X $\pm$ Y.Y]	[X.X $\pm$ Y.Y]
P1 + fine-tune	[X.X $\pm$ Y.Y]	[X.X $\pm$ Y.Y]	[X.X $\pm$ Y.Y]	[X.X $\pm$ Y.Y]
P2 full stack	[X.X $\pm$ Y.Y]	[X.X $\pm$ Y.Y]	[X.X $\pm$ Y.Y]	[X.X $\pm$ Y.Y]

Table 4: Single-feature ablations from the full stack (P2), seed 42.

Ablation	Pooled EER %	Δ
P2 (all features)	[X.X]	0
– fine-tune	[X.X]	+ [Y.Y]
– LLRD	[X.X]	+ [Y.Y]
– lang-aux	[X.X]	+ [Y.Y]
– AS-Norm	[X.X]	+ [Y.Y]
– PLDA	[X.X]	+ [Y.Y]

5.4 Headline results

5.5 Single-feature ablations

6 Discussion

6.1 Modularity in practice

[NOTE: discuss whether the strict default-off invariant held in practice across the protocol; identify any place where a feature’s behaviour leaked into the default path; reflect on the trade-off between modularity and absolute SOTA performance.]

6.2 Heterogeneous GPU utilisation efficiency

[NOTE: report wall-clock utilisation and idle-time fraction per machine; recommend allocation revisions if a particular workload turned out to be limited by something other than VRAM/compute.]

6.3 Limitations

[NOTE: benchmark v0 is read speech, near single-session, and closed-set for the fine-tuned systems (optimistic absolute EERs; benchmark v1 / SLCeleb provides the open-set in-the-wild condition); single-deployment generalisation; absence of self-supervised continued pretraining (FEATURE-009 is infrastructure-ready but not exercised here).]

7 Conclusion

We described and validated a modular, multi-GPU training pipeline for low-resource cross-lingual speaker verification. The pipeline’s eleven independently-toggable features compose cleanly under default-off invariants and ablate cleanly under single-feature negations. The reference deployment on Sinhala–Tamil bilingual audio demonstrates the pipeline’s intended use; the open-source release positions the pipeline as the substrate for downstream low-resource SV research.

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